**Dynamic Pricing Using RL**

**The Prisoner's Dilemma, repeated**

Earlier we covered the one-shot Prisoner's Dilemma: play once, mutual defection is the dominant strategy, full stop. The repeated version changes everything: if the *same two players* face each other round after round, defection today can be punished tomorrow. That threat of future punishment is what makes cooperation rational even among purely self-interested players.

This is our bridge from Bertrand (one-shot, price collapses to cost) to the possibility of sustained above-cost pricing but only if the game repeats.

**Tit-for-Tat and why it wins**

Robert Axelrod ran a tournament where researchers submitted strategies to play repeated Prisoner's Dilemma against each other round-robin. The simplest strategy submitted Tit-for-Tat won. Its rule:

1. Cooperate on the first move
2. After that, copy whatever your opponent did last round
3. Axelrod identified four properties that made it robust:
4. Nice: never defects first
5. Retaliating: punishes defection immediately
6. Forgiving:  returns to cooperation as soon as the opponent does
7. Clear: simple enough that opponents can recognize the pattern and learn to cooperate with it

"cooperate" = price high (near collusive level), "defect" = undercut.

**The Folk Theorem**

The Folk Theorem formalizes when cooperation is sustainable in infinitely (or indefinitely) repeated games: if players value future payoffs enough (a high enough "discount factor"), the threat of future punishment can outweigh the one-time gain from cheating today. This is directly why Always-Collude is fragile in a finite, known-length tournament. If both agents know exactly when the game ends, the standard backward-induction argument says cooperation unravels from the last round backward (no point cooperating on the last round since there's no future to punish you in, so no point cooperating on the second-to-last round either, and so on

**The four agents that needs to be built**

| Agent | Rule |
|---|---|
| **Always-Nash** | Always price at marginal cost c |
| **Always-Collude** | Always price at some fixed high price (e.g., monopoly price) regardless of opponent |

| | |
|---|---|
| **Tit-for-Tat** | First round: price at the collusive level. After that: mirror the opponent's *previous* price |
| **Random** | Pick a uniformly random price each round, from a **seeded** RNG |

All four agents share one job: given some history, return a price. That shared contract is what lets us swap any agent into the tournament loop interchangeably including Q-learning agent. This is the same idea as Gymnasium's standardized step() interface, just applied to agents instead of environments.

**Why each piece exists**

**price_grid stored on the agent**: Always-Nash needs to know "which index corresponds to $c$?" and Always-Collude needs "which index corresponds to the high price?" Rather than hardcoding indices, agents look these up from the grid at construction time.

**reset()**: Tit-for-Tat needs internal memory (what did the opponent do last round?), and that memory must be wiped at the start of every new tournament run

**act(history)**: takes the *shared* history because Tit-for-Tat specifically needs to see the *opponent's* last move. Random and Always-X agents will simply ignore the history argument but they all need to accept it, so the tournament loop can call every agent identically without special-casing.

Tit-for-Tat's "cooperate" price needs to be some specific collusive price but what counts as "the collusive price" isn't given to you for free the way the Nash price was (Nash price = marginal cost, trivially). A common choice is the monopoly price: the price a single firm would charge if it had no competitor at all, maximizing (p-c)(a-bp) alone.

**Monopoly price derivation**

A monopolist facing demand Q = a - bP chooses P to maximize profit:

$$\pi(P) = (P - c)(a - bP)$$

Expanding and taking the derivative with respect to P and setting it to zero, we get

$$\pi(P) = aP - bP^2 - ac + bcP$$

$$\frac{d\pi}{dP} = a - 2bP + bc = 0$$

Solving for P:

$$P = \frac{a + bc}{2b}$$

With environment's defaults (a=100, b=1, c=10):  P* = (100 + 10)/2 = 55.

Sanity check against Bertrand: the monopoly price (55) sits well above the Nash price (10), exactly the gap your Tit-for-Tat and Always-Collude agents are designed to exploit, and exactly the gap the Calvano paper found RL agents discovering on their own.

Tit-for-Tat gets exploited once (round 0: it prices high at 55, AlwaysNash undercuts to 10, TFT earns zero that round since it's priced out of the market), then immediately punishes by collapsing to match AlwaysNash's price of 10 for every subsequent round. From round 1 onward, both agents are stuck at the Bertrand price, both earning zero profit. This is the Folk Theorem playing out in miniature: cooperation only survives against an opponent willing to reciprocate it, and Always-Nash never will, so TFT correctly settles into permanent defection against it.

Against a partner who's also "nice," Tit-for-Tat sustains the **monopoly price** every round, with both agents earning the maximum possible profit (1012 each, split evenly) way above the 0 they'd earn at the Bertrand price.